

Problem A. Easy Linear Programming

Time Limit 2000 ms

Problem Statement

We have A cards, each of which has an integer 1 written on it. Similarly, we also have B cards with 0s and C cards with -1 s.

We will pick up K among these cards. What is the maximum possible sum of the numbers written on the cards chosen?

Constraints

- All values in input are integers.
- $0 \leq A, B, C$
- $1 \leq K \leq A + B + C \leq 2 \times 10^9$

Input

Input is given from Standard Input in the following format:

$A \ B \ C \ K$

Output

Print the maximum possible sum of the numbers written on the cards chosen.

Sample 1

Input	Output
2 1 1 3	2

Consider picking up two cards with 1s and one card with a 0. In this case, the sum of the numbers written on the cards is 2, which is the maximum possible value.

Sample 2

Input	Output
1 2 3 4	0

Sample 3

Input	Output
20000000000 0 0 20000000000	20000000000

Problem B. Four Hidden

Time Limit 2000 ms

Problem Statement

You are given a string T consisting of lowercase English letters and `?`, and a string U consisting of lowercase English letters.

The string T is obtained by taking some lowercase-only string S and replacing exactly four of its characters with `?`.

Determine whether it is possible that the original string S contained U as a contiguous substring.

Constraints

- T is a string of length between 4 and 10, inclusive, consisting of lowercase letters and `?`.
- T contains exactly four occurrences of `?`.
- U is a string of length between 1 and $|T|$, inclusive, consisting of lowercase letters.

Input

The input is given from Standard Input in the following format:

```
 $T$ 
 $U$ 
```

Output

Print `Yes` if it is possible that the original string S contained U as a contiguous substring; otherwise, print `No`.

Sample 1

Input	Output
tak??a?h? nashi	Yes

For example, if S is **takanashi**, it contains **nashi** as a contiguous substring.

Sample 2

Input	Output
??e??e snuke	No

No matter what characters replace the **?** s in T , S cannot contain **snuke** as a contiguous substring.

Sample 3

Input	Output
???? aoki	Yes

Problem C. 1 2 1 3 1 2 1

Time Limit 2000 ms

Problem Statement

We define sequences S_n as follows.

- S_1 is a sequence of length 1 containing a single 1.
- S_n (n is an integer greater than or equal to 2) is a sequence obtained by concatenating S_{n-1} , n , S_{n-1} in this order.

For example, S_2 and S_3 is defined as follows.

- S_2 is a concatenation of S_1 , 2, and S_1 , in this order, so it is 1, 2, 1.
- S_3 is a concatenation of S_2 , 3, and S_2 , in this order, so it is 1, 2, 1, 3, 1, 2, 1.

Given N , print the entire sequence S_N .

Constraints

- N is an integer.
- $1 \leq N \leq 16$

Input

Input is given from Standard Input in the following format:

N

Output

Print S_N , with spaces in between.

Sample 1

Input	Output
2	1 2 1

As described in the Problem Statement, S_2 is 1, 2, 1.

Sample 2

Input	Output
1	1

Sample 3

Input	Output
4	1 2 1 3 1 2 1 4 1 2 1 3 1 2 1

- S_4 is a concatenation of S_3 , 4, and S_3 , in this order.

Problem D. Creating Strings

Time Limit 1000 ms

Mem Limit 524288 kB

Given a string, your task is to generate all different strings that can be created using its characters.

Input

The only input line has a string of length n . Each character is between a–z.

Output

First print an integer k : the number of strings. Then print k lines: the strings in alphabetical order.

Constraints

- $1 \leq n \leq 8$

Example

Input	Output
aabac	20 aaabc aaacb aabac aabca aacab aacba abaac abaca abcaa acaab acaba acbaa baaac baaca baca bcaaa caaab caaba cabaa cbaaa

Problem E. Perfect String

Time Limit 2000 ms

Problem Statement

Let us call a string consisting of uppercase and lowercase English alphabets a **wonderful string** if all of the following conditions are satisfied:

- The string contains an uppercase English alphabet.
- The string contains a lowercase English alphabet.
- All characters in the string are pairwise distinct.

For example, **AtCoder** and **Aa** are wonderful strings, while **atcoder** and **Perfect** are not.

Given a string S , determine if S is a wonderful string.

Constraints

- $1 \leq |S| \leq 100$
- S is a string consisting of uppercase and lowercase English alphabets.

Input

Input is given from Standard Input in the following format:

S

Output

If S is a wonderful string, print **Yes**; otherwise, print **No**.

Sample 1

Input	Output
AtCoder	Yes

AtCoder is a wonderful string because it contains an uppercase English alphabet, a lowercase English alphabet, and all characters in the string are pairwise distinct.

Sample 2

Input	Output
Aa	Yes

Note that **A** and **a** are different characters. This string is a wonderful string.

Sample 3

Input	Output
atcoder	No

It is not a wonderful string because it does not contain an uppercase English alphabet.

Sample 4

Input	Output
Perfect	No

It is not a wonderful string because the 2-nd and the 5-th characters are the same.

Problem F. To 3

Time Limit 2000 ms

Problem Statement

Given is a positive integer N , where none of the digits is 0.

Let k be the number of digits in N . We want to make a multiple of 3 by erasing at least 0 and at most $k - 1$ digits from N and concatenating the remaining digits without changing the order.

Determine whether it is possible to make a multiple of 3 in this way. If it is possible, find the minimum number of digits that must be erased to make such a number.

Constraints

- $1 \leq N < 10^{18}$
- None of the digits in N is 0.

Input

Input is given from Standard Input in the following format:

N

Output

If it is impossible to make a multiple of 3, print **-1**; otherwise, print the minimum number of digits that must be erased to make such a number.

Sample 1

Input	Output
35	1

By erasing the 5, we get the number 3, which is a multiple of 3. Here we erased the minimum possible number of digits - 1.

Sample 2

Input	Output
369	0

Note that we can choose to erase no digit.

Sample 3

Input	Output
6227384	1

For example, by erasing the 8, we get the number 622734, which is a multiple of 3.

Sample 4

Input	Output
11	-1

Note that we must erase at least 0 and at most $k - 1$ digits, where k is the number of digits in N , so we cannot erase all the digits.

In this case, it is impossible to make a multiple of 3 in the way described in the problem statement, so we should print **-1**.

Problem G. Almost Equal

Time Limit 2000 ms

Problem Statement

You are given N strings $S_1, S_2, \dots, S_N$, each of length M , consisting of lowercase English letter. Here, S_i are pairwise distinct.

Determine if one can rearrange these strings to obtain a new sequence of strings $T_1, T_2, \dots, T_N$ such that:

- for all integers i such that $1 \leq i \leq N - 1$, one can alter exactly one character of T_i to another lowercase English letter to make it equal to T_{i+1} .

Constraints

- $2 \leq N \leq 8$
- $1 \leq M \leq 5$
- S_i is a string of length M consisting of lowercase English letters. ($1 \leq i \leq N$)
- S_i are pairwise distinct.

Input

The input is given from Standard Input in the following format:

```
N M
S1
S2
⋮
SN
```

Output

Print **Yes** if one can obtain a conforming sequence; print **No** otherwise.

Sample 1

Input	Output
4 4 bbed abcd abed fbed	Yes

One can rearrange them in this order: **abcd** , **abed** , **bbed** , **fbed** . This sequence satisfies the condition.

Sample 2

Input	Output
2 5 abcde abced	No

No matter how the strings are rearranged, the condition is never satisfied.

Sample 3

Input	Output
8 4 fast face cast race fact rice nice case	Yes

Problem H. Common Raccoon vs Monster

Time Limit 2000 ms

Problem Statement

Raccoon is fighting with a monster.

The *health* of the monster is H .

Raccoon can use N kinds of special moves. Using the i -th move decreases the monster's health by A_i . There is no other way to decrease the monster's health.

Raccoon wins when the monster's health becomes 0 or below.

If Raccoon can win without using the same move twice or more, print **Yes**; otherwise, print **No**.

Constraints

- $1 \leq H \leq 10^9$
- $1 \leq N \leq 10^5$
- $1 \leq A_i \leq 10^4$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
H N  
A1 A2 ... AN
```

Output

If Raccoon can win without using the same move twice or more, print **Yes**; otherwise, print **No**.

Sample 1

Input	Output
10 3 4 5 6	Yes

The monster's health will become 0 or below after, for example, using the second and third moves.

Sample 2

Input	Output
20 3 4 5 6	No

Sample 3

Input	Output
210 5 31 41 59 26 53	Yes

Sample 4

Input	Output
211 5 31 41 59 26 53	No

Problem I. Tower of Hanoi

Time Limit 1000 ms

Mem Limit 524288 kB

The Tower of Hanoi game consists of three stacks (left, middle and right) and n round disks of different sizes. Initially, the left stack has all the disks, in increasing order of size from top to bottom.

The goal is to move all the disks to the right stack using the middle stack. On each move you can move the uppermost disk from a stack to another stack. In addition, it is not allowed to place a larger disk on a smaller disk.

Your task is to find a solution that minimizes the number of moves.

Input

The only input line has an integer n : the number of disks.

Output

First print an integer k : the minimum number of moves.

After this, print k lines that describe the moves. Each line has two integers a and b : you move a disk from stack a to stack b .

Constraints

- $1 \leq n \leq 16$

Example

Input	Output
2	3 1 2 1 3 2 3

Problem J. Apple Division

Time Limit 1000 ms

Mem Limit 524288 kB

There are n apples with known weights. Your task is to divide the apples into two groups so that the difference between the weights of the groups is minimal.

Input

The first input line has an integer n : the number of apples.

The next line has n integers $p_1, p_2, \dots, p_n$: the weight of each apple.

Output

Print one integer: the minimum difference between the weights of the groups.

Constraints

- $1 \leq n \leq 20$
- $1 \leq p_i \leq 10^9$

Example

Input	Output
5 3 2 7 4 1	1

Explanation: Group 1 has weights 2, 3 and 4 (total weight 9), and group 2 has weights 1 and 7 (total weight 8).